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“Ecophotonics”: Turning Nature’s Light Management Strategies into Sustainable Technology

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Sustainable & Bioinspired Materials

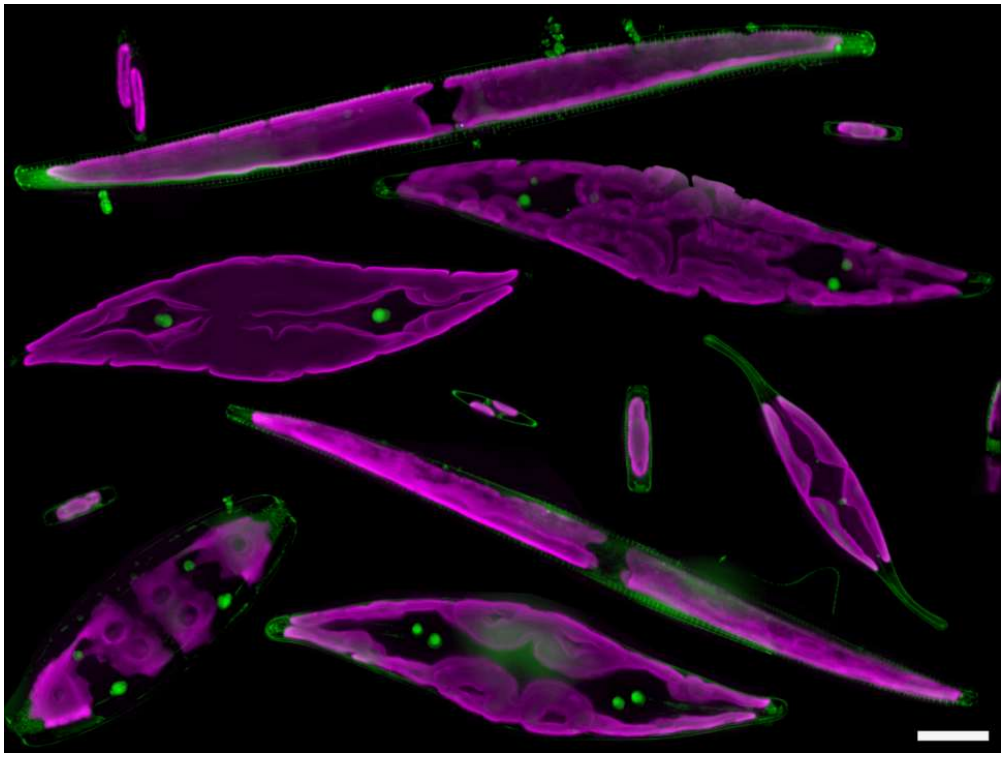
- **Dr. Vera Titze launches Ecophotonics:** Thanks to the Max Planck Society’s Minerva Fast-Track Program, she is setting up her independent research group in the Department of Sustainable and Bio-Inspired Material.
- **How nature manages light:** Ecophotonics will explore how living systems process light and how living materials based on bacteria or microalgae can carry out light-based functionalities.
- **Impact beyond the lab:** Taking inspiration from nature, the group will develop new bio-hybrid photonic materials and optical devices, including bio-derived lasers and self-powered environmental sensors.

Dr. Vera Titze wants to turn nature’s light-management strategies into bio-based optical materials for low-power optical signal technologies. The Max Planck Society is backing her vision with a Minerva Fast-Track Fellowship, a highly selective program that awards six fellowships each year to outstanding women as they establish an independent research profile. In March, Titze will launch Ecophotonics in the Department of Sustainable and Bio-Inspired Materials and recruit one PhD student and one postdoc.

The name Ecophotonics captures Titze’s curiosity for fundamental science and her commitment to impact beyond the lab. Her team will zoom in on the adaptive mechanisms that living systems use across scales to manipulate light and will translate these insights into responsive optical materials and devices. “*The Department of Sustainable and Bio-Inspired Materials is an ideal place for my group because it encourages truly interdisciplinary work. Here, we can connect optical technologies with applications that benefit the environment,*” she says.

Understanding how Bacteria and Microalgae Detect and Manage Light

At the heart of Ecophotonics is a simple premise: living systems can teach us how to use light as a rich carrier of information.



Marine architects of light: A fluorescence microscopy collage of diatoms, showcasing their remarkable geometry.

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> "*Light processing mechanisms in nature have been optimized through millions of years of evolution*," Titze explains. "*What fascinates me is how nature solves complex challenges with low energy consumption and with materials entirely from sustainable building blocks*."

Some microalgae and bacteria, for instance, detect changes in light and respond by adjusting how they grow and how they manage energy — and, in some cases, by producing optical signals of their own.

The Ecophotonics team will study optical information processing in diatoms, single-celled microalgae that are masters at adapting to rapidly shifting conditions. Studying how individual cells manipulate light and how cell communities interact can provide the basis for adaptive, living optical materials. The group is already building specialized microscopes to trace how light propagates through living matter.

Another focus will be on the striking order that emerges when marine bacteria like ***Flavobacteria*** grow together. They can form colonies that arrange themselves into periodic structures known as photonic crystals, making them attractive for high-performance optical applications. By studying how these photonic

crystals grow, the team learns how to engineer the properties of sustainable bacteria-based materials.

From Living Structures to Low-Power Sensors and Optical Devices

"*Once we understand these fundamental design principles*, *we can envision real-world solutions entirely from living matter,*" Titze adds.

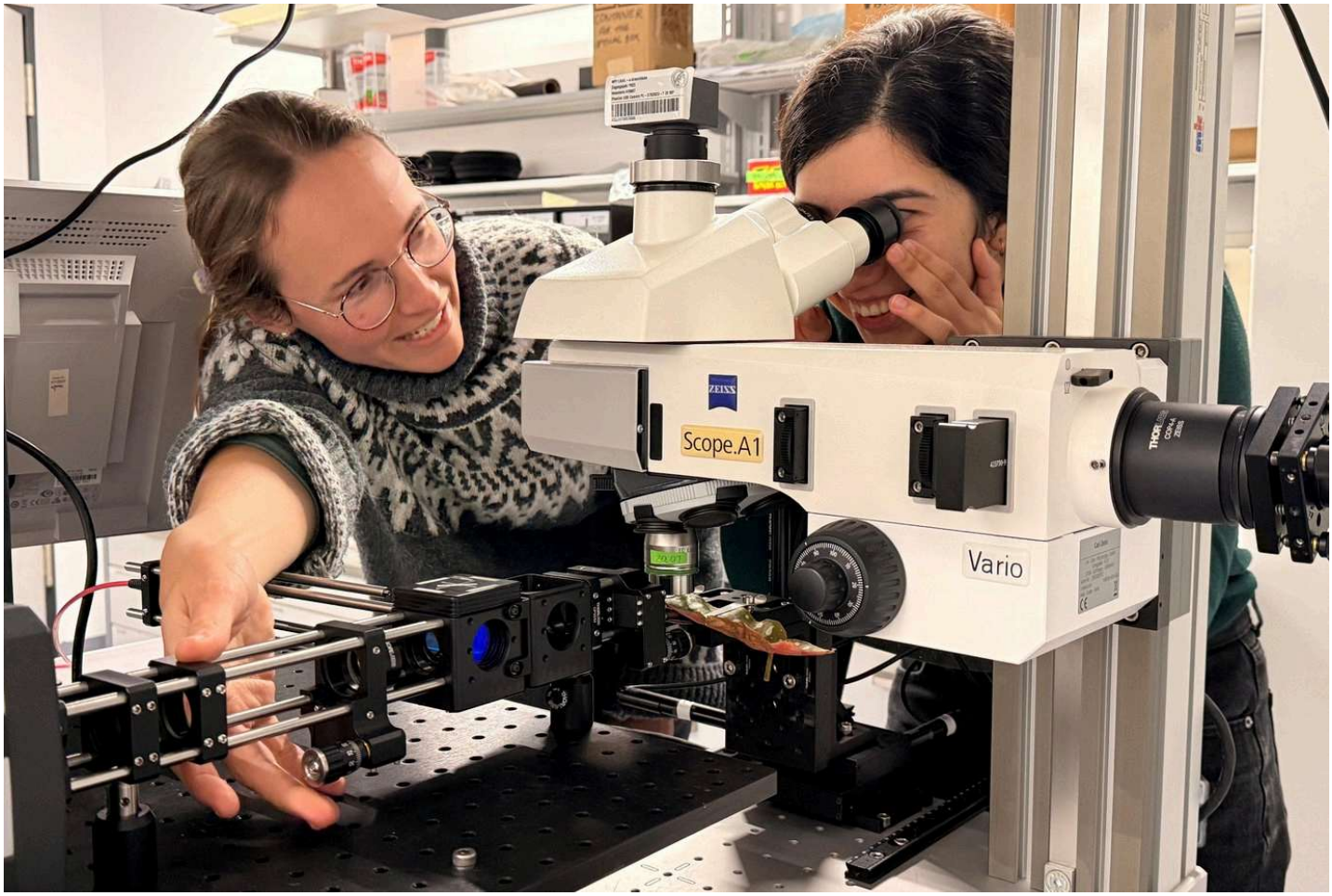
One possible application is to use bacterial colonies to build optical devices. Instead of manufacturing components piece by piece from conventional materials, bacteria can be cultivated with simple nutrients and grown into programmable patterns that shape how light interacts with them. In collaboration with biotechnologists in the Department, the group works with bacteria tailored to emit light, opening new possibilities for adaptable, self-growing light sources.

Another potential real-world application reflects Titze's commitment to sustainability.

> "*Nowadays, we monitor environmental changes with devices that need constant power to measure, store, and transmit data. Imagine using living materials to turn those changes into a smart light signal instead*," she says.

Such low-power sensors could integrate biological components or mimic their tricks to save energy, for instance, by shifting color or brightness in response to changing conditions.

As Ecophotonics takes shape at the interface of physics, microbiology, and materials science, Titze **welcomes** motivated Bachelor's and Master's students to get in touch about opportunities to contribute to the group's research.



Bright ideas in progress: Dr. Vera Titze and student Ana Karina Riaño assemble a custom microscope to track light in living matter.

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